

Gate R1, discharged: the cutoff-uniformity of the Glimm–Jaffe–Spencer estimates

Abstract

Gate R1 asked whether the constant in [1, Thm. 1.1.8] is uniform over the full cutoff class $\{h : 0 \leq h \leq 1, \text{supp } h \text{ compact}\}$, the printed statement saying only “uniformly as $h \rightarrow 1$ ”. The gate is **discharged**, on three independent grounds.

(1) The uniformity is printed after all. On p. 629, in §4 of [1]: “*The estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff h , and the path space integrals converge as $h \rightarrow 1$.*” This sentence, missed in the first audit pass because it sits in §4 rather than §1.1, asserts class-uniformity for precisely the four theorems the programme uses.

(2) The mechanism audit explains why. The convergence proof is a *kernel-norm* argument: the norm (2.4.3) contains no Gaussian integral and no e^{-V} . The cutoff enters the expansion in exactly two ways — multiplicatively inside vertex kernels, where it is majorised at the first step by $|\lambda h(x)| \leq \lambda$ (printed: “ $v(x)$ is $O(\lambda)$ times the characteristic function of a lattice square”, p. 626), and through the spectator factor $e^{-V(h)}$, which is never estimated: the expansion terminates in terms proportional to $\int e^{-V} d\Phi$, and the final division by exactly this quantity cancels every occurrence (their (4.3)–(4.4)). Hence no stability bound enters any constant, and the constants are functions of $(m_0, K_2, K_1, \lambda, n, \varepsilon, \mathcal{P})$ only — matching the printed constant-choice order of Theorem 2.4.1.

(3) The only h -sensitive analytic prerequisites are proved here, uniformly. Well-definedness of dq_h and the integrability that legitimises the expansion identities require two facts, both established below with all h -dependence explicit: $e^{-\lambda|a_0||K|} \leq Z(h) < \infty$ (Lemma 4.1) and $\sup_{0 \leq h \leq 1, \text{supp } h \subseteq K} \|e^{-V(h)}\|_{L^p(d\Phi)} < \infty$ (Theorem 4.4). The two properties used are exactly $h \geq 0$ (in the pointwise Wick lower bound) and $h \leq 1$ (in the L^2 kernel bounds) — explaining why GJS chose the class $0 \leq h \leq 1$ on p. 589.

Consequently the (M_2) theorem of `m2-audit.tex` and the (G) theorem of `g-side.tex` are conditional on nothing beyond the printed theorems of [1] themselves; the weak-coupling closure holds at the ordinary standard of citation. Gate R2 (the K_1 exponent) remains open but affects only the numerical size of constants.

1 The gate

Definition 1.1 (R1). Let C_8 be the constant of [1, Thm. 1.1.8]. R1 is the assertion that C_8 may be chosen independently of h within the class $\mathcal{C} := \{h \text{ measurable} : 0 \leq h \leq 1, \text{supp } h \text{ compact}\}$ of [1, p. 589] — not merely along some sequence $h \uparrow 1$. The derivations of (M_2) and (G) use members $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$ of $\mathcal{C}$ which do *not* increase to 1, so R1 is load-bearing.

2 The printed statement

Source Fact 2.1 (p. 629, §4). “*It is important to work with a translation invariant Hamiltonian, in order that time translation preserve subspaces of bounded momentum.* **The**

estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff h , and the path space integrals converge as $h \rightarrow 1$. Thus these theorems hold in the infinite volume limit, and we proceed to work in the limit $h = 1$.

Two remarks on the reading. First, the sentence separates the two assertions — uniformity of the estimates in h , and convergence as $h \rightarrow 1$ — so “uniform in h ” quantifies over the class, not over a tail of a sequence; the second clause would otherwise be redundant. Second, the four theorems named are exactly the moment bound (1.1.8), the decoupling estimate (1.1.11), and the two norm-decrease engines (3.1, 4.1); together with Theorem 1.1.7 (printed “independent of h ” in its own statement, p. 594) these are all the source inputs of `m2-audit.tex` and `g-side.tex`.

This alone would settle R1 at the citation standard. The remainder of this note establishes the stronger claim: the uniformity is *structural*, i.e. visible in the anatomy of the proof, so that the reading does not rest on one sentence.

3 Mechanism audit

The expansion machinery of [1, §§2–4] manipulates integrals $\int R(\Phi) B e^{-V(h)} d\Phi_C$, C an interpolated covariance, through five devices. We list every point of contact with h .

Mechanism 3.1 (the norms are kernel norms; no h). The norm (2.4.3) of an expansion term m_α is $\int du ds \exp[\cdot \cdot] \|w(\alpha, u, s, y, \cdot)\|_{L^2}$ times combinatorial weights (2.4.4)–(2.4.5) built from $K_1, K_2, N(\Delta)$, times $e^{\tau t}$: a functional of the *kernel* w alone. No Gaussian expectation, no e^{-V} , no h appears in the norm. The norm-decrease Theorems 2.4.1, 3.1, 4.1 are inequalities between such kernel functionals, and their proofs (Lemmas 2.4.2–2.4.4, (3.2)–(3.6), §4) estimate kernels: contraction kernels δC against vertex kernels.

Mechanism 3.2 (covariances are h -free). The interpolation $C(s) = sC' + (1-s)C''$ (2.0.1) is between free and Dirichlet-on-contours covariances; the kernels (2.2.2)–(2.2.3) are built from $(-d^2/dx^2 + m_0^2)$ and contour geometry. Verified legible: no h enters any covariance, any contraction kernel, or any of the Propositions 2.1.x.

Mechanism 3.3 (h in vertices, majorised at first contact). Vertices arise from functional derivatives of the spectator:

$$\frac{\delta}{\delta \Phi(x)} e^{-V(h)} = -\lambda h(x) : \mathcal{P}'(\Phi(x)) : e^{-V(h)},$$

and iterates with $\mathcal{P}'', \dots$. So a vertex contributes to the *kernel* a factor $\lambda h(x) p_j$ supported in a lattice square. The very first estimate applied to it is the printed one (p. 626): “ $v(x)$ is $O(\lambda)$ times the characteristic function of a lattice square” — i.e. $|\lambda h(x) p_j| \leq O(\lambda) \chi_\Delta(x)$, using $0 \leq h \leq 1$ and nothing else. Downstream of this line the kernels are h -free. This is the single point where h touches a quantitative estimate.

Mechanism 3.4 (the identities are exact and h -blind). The devices (2.0.4)–(2.0.5) (covariance interpolation), (2.2.1), (2.2.5), (2.2.7) (advancing annihilators), (4.1)–(4.2), (4.5) (integration by parts) are identities applied to integrands $F = R(\Phi) B_T e^{-V(h)}$; $e^{-V(h)}$ rides along inside F . Their validity requires only that the integrals exist — integrability, not bounds. Theorem 4.4 below supplies this uniformly over $\mathcal{C}$ (per- h finiteness would suffice).

Mechanism 3.5 (termination cancels every e^{-V} ; no stability constant). The full expansion (4.3) reads

$$\int AB_T e^{-V} d\Phi_{C_0} = \sum_i \int \left[\int dy w_i(y) : \Phi(y_1) \cdots \Phi(y_i) : \right] B_T e^{-V} d\Phi_{C_0} + \sum_r w_r \int e^{-V} d\Phi_{C_0},$$

with w_i, w_r numerical kernels controlled by the norm inequalities; the remainder terms are literally multiples of $\int e^{-V} d\Phi$ (Table 1: final- r terms have “no vertices, no annihilators”). The proof of Theorems 1.1.8/1.1.11 then *divides (4.3) by $\int e^{-V} d\Phi_{C_0}$* (p. 628): the first sum reassembles into dq_h -integrals of the comparison polynomials, and the remainder becomes $\sum_r w_r$ exactly. Every factor $e^{-V(h)}$ cancels against the normalisation. Consequently *no upper or lower bound on $\int e^{-V} d\Phi$, and no stability estimate, enters any constant of the proof.*

Proposition 3.6 (structural uniformity). *In the proofs of Theorems 1.1.7, 1.1.8, 1.1.11, 2.4.1, 3.1 and 4.1 of [1], the cutoff h enters only through Mechanism 3.3, where it is majorised using $0 \leq h \leq 1$, and through Mechanism 3.4, where only existence of the integrals is used. All constants are determined by $(m_0, K_2, K_1, \lambda, n, \varepsilon, \mathcal{P})$, chosen in the printed order (proof of Thm. 2.4.1: “We choose the constants of our theorem in the following order: m_0, K_2, K_1 , and then λ ”), with no reference to h .*

Proof. By Mechanisms 3.1–3.5: the quantitative chain is entirely at the level of kernel norms (3.1), whose ingredients are h -free covariance/contraction kernels (3.2) and vertex kernels majorised by $O(\lambda)\chi_\Delta$ (3.3); the measure-level steps are exact identities (3.4) or exact cancellations (3.5). $\square$

4 The two h -sensitive analytic facts, proved

Notation: $d\Phi$ = Gaussian with covariance $G = (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$; $G(\zeta) = \frac{1}{2\pi} K_0(m_0|\zeta|)$; $\mathcal{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$; $h \in \mathcal{C}$ with $\text{supp } h \subseteq K$, K compact; $V(h) = \lambda \int h : \mathcal{P}(\Phi) : dz$, $Z(h) = \int e^{-V(h)} d\Phi$. Let δ_κ be a fixed smooth approximate identity at scale κ^{-1} , $\Phi_\kappa = \delta_\kappa * \Phi$, $c_\kappa(z) = \mathbb{E}[\Phi_\kappa(z)^2] \leq \frac{1}{2\pi} \log \kappa + C_0$ (translation invariance), and $V_\kappa(h) = \lambda \int h : \mathcal{P}(\Phi_\kappa) :_{c_\kappa} dz$, Wick ordered with respect to the variance of the cutoff field, so that $V_\kappa(h) \rightarrow V(h)$ in every $L^p(d\Phi)$.

Lemma 4.1 (normalisation). *For every $h \in \mathcal{C}$ with $\text{supp } h \subseteq K$: $Z(h) \geq e^{-\lambda|a_0||K|} > 0$, and $Z(h) < \infty$ by Theorem 4.4. Hence dq_h is a well-defined probability measure on the whole class $\mathcal{C}$.*

Proof. $\mathbb{E}[V(h)] = \lambda a_0 \int h \in [-\lambda|a_0||K|, \lambda|a_0||K|]$, since Wick monomials of degree ≥ 1 have mean zero and $0 \leq h \leq \mathbf{1}_K$. Jensen’s inequality for the probability measure $d\Phi$ gives $Z(h) \geq e^{-\mathbb{E}[V(h)]} \geq e^{-\lambda|a_0||K|}$. $\square$

Lemma 4.2 (Wick lower bound; the $h \geq 0$ entry). *There is $b = b(\mathcal{P})$ such that for all $c \geq 0$ and $x \in \mathbb{R}$, $:\mathcal{P}:_c(x) := \sum_n a_n c^{n/2} \text{He}_n(x/\sqrt{c}) \geq -b(1+c)^{p'}$. Consequently, for every $h \in \mathcal{C}$ with $\text{supp } h \subseteq K$ and every $\kappa \geq 2$,*

$$V_\kappa(h) \geq -\lambda b |K| \left(1 + \frac{1}{2\pi} \log \kappa + C_0\right)^{p'} \geq -c_1(\lambda, \mathcal{P}, K) (\log \kappa)^{p'}.$$

Proof. Expanding the Hermite polynomials, $:\mathcal{P}:_c(x) = \sum_{m=0}^{2p'} \beta_m x^m$ with $\beta_{2p'} = a_{2p'}$ and $|\beta_m| \leq C(\mathcal{P})(1+c)^{(2p'-m)/2}$ (each power c^j comes with x^{n-2j}). For a polynomial

$Ax^{2p'} + \sum_{m < 2p'} \beta_m x^m$ with $A > 0$, weighted AM–GM gives, for each $m < 2p'$,

$$|\beta_m| |x|^m \leq \frac{A}{4p'} x^{2p'} + C_{p'} \left(\frac{|\beta_m|^{2p'}}{A^m} \right)^{1/(2p'-m)},$$

and summing, $\min_x : \mathcal{P} :_c \geq -C'_{p'} \sum_m (|\beta_m|^{2p'}/A^m)^{1/(2p'-m)}$. With $|\beta_m| \leq C(1+c)^{(2p'-m)/2}$ the exponent of $(1+c)$ is $\frac{2p'}{2p'-m} \cdot \frac{2p'-m}{2} = p'$, uniformly in m . The second claim follows from $0 \leq h$, $\int h \leq |K|$, and the pointwise bound applied under the integral — this is the single use of the *sign* of h . $\square$

Lemma 4.3 (ultraviolet difference; the $h \leq 1$ entry). *There is $c_2 = c_2(\lambda, \mathcal{P}, K)$ such that for all $h \in \mathcal{C}$ with $\text{supp } h \subseteq K$ and all $\kappa \geq 2$,*

$$\|V(h) - V_\kappa(h)\|_{L^2(d\Phi)}^2 \leq c_2 \kappa^{-1} (\log \kappa)^{2p'}.$$

Proof. By the Wick isometry, with $G_\kappa^{(1)} := (\delta_\kappa \otimes 1) * G$ and $G_\kappa := (\delta_\kappa \otimes \delta_\kappa) * G$,

$$\|V(h) - V_\kappa(h)\|_{L^2}^2 = \lambda^2 \sum_{n=1}^{2p'} a_n^2 n! \iint h(z) h(z') [G^n - 2(G_\kappa^{(1)})^n + G_\kappa^n] (z - z') dz dz',$$

(the c_κ -Wick ordering makes the diagonal subtractions match; degree 0 cancels). Using $|h| \leq \mathbf{1}_K$ — the single use of the *upper* bound on h — it suffices to bound $\iint_{K \times K} |G^n - (J)^n|$ for $J \in \{G_\kappa^{(1)}, G_\kappa\}$. The classical Bessel facts $0 < G(\zeta) \leq C(1 + |\log |\zeta||) e^{-m_0|\zeta|}$ and $|\nabla G(\zeta)| \leq C(|\zeta|^{-1} + 1) e^{-m_0|\zeta|}$ give, for the average J of G over balls of radius $\leq 2/\kappa$: $J(\zeta) \leq C(1 + \log \kappa)$ for all ζ , and $|J(\zeta) - G(\zeta)| \leq C\kappa^{-1}(|\zeta|^{-1} + 1)$ for $|\zeta| \geq 4/\kappa$. Split the integral over $\zeta = z - z'$:

$$\int_{|\zeta| < 4/\kappa} [G^n + J^n] \leq C_n \left[\int_{|\zeta| < 4/\kappa} (1 + |\log |\zeta||)^n d\zeta + \kappa^{-2} (1 + \log \kappa)^n \right] \leq C'_n \kappa^{-2} (\log \kappa)^n,$$

$$\int_{4/\kappa \leq |\zeta| \leq R} n \max(G, J)^{n-1} |G - J| \leq C_n \kappa^{-1} \int_0^R (1 + |\log r|)^{n-1} (1 + r) dr \leq C_n(R) \kappa^{-1},$$

with $R = \text{diam } K$; multiply by $|K|$ for the outer variable and sum over $n \leq 2p'$. $\square$

Theorem 4.4 (h -uniform stability). *For every $p < \infty$ and compact K ,*

$$\sup \{ \|e^{-V(h)}\|_{L^p(d\Phi)} : h \in \mathcal{C}, \text{supp } h \subseteq K \} < \infty.$$

Proof. $V(h) - V_\kappa(h)$ is a sum of Wick chaoses of order $\leq 2p'$, so Nelson's hypercontractive moment comparison ($\|X\|_{L^q} \leq (q-1)^{r/2} \|X\|_{L^2}$ for X in the r -th chaos; an h -free tool, [2, §I.5, Thm. V.9]) applies. For $b \geq b_0(\lambda, \mathcal{P}, K)$ choose $\kappa = \kappa(b)$ by $c_1(\log \kappa)^{p'} = b/2$, i.e. $L := \log \kappa = (b/2c_1)^{1/p'}$. By Lemma 4.2, $\{V(h) \leq -b\} \subseteq \{V(h) - V_\kappa(h) \leq -b/2\}$, so by Chebyshev in L^{2q} and Lemma 4.3,

$$\Pr[V(h) \leq -b] \leq \left(\frac{2}{b}\right)^{2q} (2q)^{2p'q} (c_2 e^{-L} L^{2p'})^q = \left[\frac{2\sqrt{c_2}}{b} (2q)^{p'} L^{p'} e^{-L/2}\right]^{2q}.$$

Choose $2q = \exp(\frac{L}{4p'})$, so that $(2q)^{p'} = e^{L/4}$ and the bracket is $\leq \frac{2\sqrt{c_2}}{b} L^{p'} e^{-L/2} \leq e^{-1}$ for $b \geq b_1(\lambda, \mathcal{P}, K)$, whence

$$\Pr[V(h) \leq -b] \leq \exp\left(-\exp\left(\frac{1}{4p'}(b/2c_1)^{1/p'}\right)\right),$$

doubly exponentially small, uniformly over the stated class; and $\exp(\frac{1}{4p'}(b/2c_1)^{1/p'}) \geq 2pb$ for b large, since a power of b inside an exponential beats $\log b$. Then $\|e^{-V(h)}\|_{L^p}^p = \int_0^\infty pe^{pb} \Pr[V(h) \leq -b] db + O(e^{pb_1}) < \infty$ with a bound depending only on $(p, \lambda, \mathcal{P}, K)$. Every constant above came from Lemmas 4.2–4.3, which used exactly $h \geq 0$ and $h \leq 1$. $\square$

Remark 4.5 (why the class is what it is). The two halves of the constraint $0 \leq h \leq 1$ are used in complementary places: positivity in the pointwise lower bound (Lemma 4.2 would fail for h of variable sign — the interaction density is unbounded above *and* below), boundedness in the L^2 kernel estimates (Lemma 4.3 and Mechanism 3.3). Nothing else about h — smoothness, shape, time extent, monotonicity along a sequence — is touched anywhere. This is the structural content of Source Fact 2.1.

5 Verdict and consequences

Theorem 5.1 (R1 discharged). *The constants of Theorems 1.1.7, 1.1.8, 1.1.11, 3.1 and 4.1 of [1] are uniform over $\mathcal{C} = \{0 \leq h \leq 1, \text{supp } h \text{ compact}\}$. Grounds: the printed Source Fact 2.1 and the printed statement of Thm. 1.1.7; the structural Proposition 3.6; and the h -uniform analytic prerequisites Lemma 4.1 and Theorem 4.4. In particular the slab family $\{h_{T,g} : T > 0, g \in \mathcal{G}\}$ is covered.*

Corollary 5.2. (i) *Theorem 4.4 of `m2-audit.tex` — the uniform local second moment (M_2) — holds conditional only on the printed [1, Thm. 1.1.8].* (ii) *Theorem 7.2 of `g-side.tex` — the uniform gap (G) with $\gamma = m_1$ — holds conditional only on the printed [1, Thms. 1.1.7, 1.1.8].* (iii) *With (G) and (M_2) supplied to the conditional LPPL theorem of `lppl.tex`, the full-net local-norm convergence of the spatial-cutoff ground states ω_g , with all its listed consequences, holds in the GJS weak-coupling region at the ordinary standard of citation of published theorems.*

Remark 5.3 (what remains open, exactly). (a) Gate R2: the exponent on K_1 in the norm (1.1.8) is illegible in the scan; it affects the numerical size of $\mathfrak{K}(p)$ in (M_2) and nothing else. (b) The OCR did not permit certifying each numeric constant inside (2.4.6)–(2.4.17), (3.2)–(3.6); by Proposition 3.6 these constants are h -free, so R1 does not depend on them — only the eventual bookkeeping of explicit numerical values does. (c) Everything remains restricted to the GJS weak-coupling region λ/m_0^2 small; no claim for the full noncritical region.

6 Status

- **Discharged:** Gate R1, by printed statement (p. 629) + mechanism audit (§3) + self-contained proofs of the only two h -sensitive prerequisites (§4).
- **Upgraded:** (M_2) and (G) are now conditional only on the printed theorems of [1]; the weak-coupling closure of the spatial-cutoff programme holds at citation standard.
- **Open:** R2 (constant size only); the numerical bookkeeping of Remark 5.3(b); and everything beyond weak coupling.

References

- [1] J. Glimm, A. Jaffe and T. Spencer, *The Wightman axioms and particle structure in the $\mathcal{P}(\phi)_2$ quantum field model*, Ann. of Math. **100** (1974) 585–632.

- [2] B. Simon, *The $P(\Phi)_2$ Euclidean (Quantum) Field Theory*, Princeton University Press, 1974.